

Java OOP - Introduction to Class & Object

1. Why OOP Exists

Java's OOP lets us represent real things (Car, Student, BankAccount) as objects with properties (data) and behaviors (methods).

It organizes code, improves reusability, and scales well for large applications.

2. Core Concepts

Class: A blueprint/template that defines data (fields) and actions (methods) an object can have.

Object: A real instance created from a class with its own values for fields.

Analogy: Class -> Recipe, Object -> Actual cake made from the recipe.

3. Syntax of a Class in Java

```
class Car {  
    String brand;  
    int speed;  
  
    void accelerate() {  
        speed += 10;  
        System.out.println("Speed: " + speed);  
    }  
}
```

4. Creating Objects

```
public class Main {  
    public static void main(String[] args) {  
        Car c1 = new Car();  
        Car c2 = new Car();  
  
        c1.brand = "Tesla";  
        c1.speed = 0;  
  
        c2.brand = "BMW";  
        c2.speed = 20;  
  
        c1.accelerate();  
        c2.accelerate();  
    }  
}
```

5. Memory Model Insight

When creating an object with 'new', Java allocates memory on the heap for its fields.

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The stack stores references, and the method area stores method definitions.

6. Real-World Example

```
class Student {
    String name;
    int age;

    void display() {
        System.out.println(name + " (" + age + ")");
    }
}

public class Main {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.name = "Ravi";
        s1.age = 20;

        Student s2 = new Student();
        s2.name = "Priya";
        s2.age = 19;

        s1.display();
        s2.display();
    }
}
```

7. Common Pitfalls

1. Null references cause NullPointerException.
2. Each object has its own copy of instance variables.
3. Static variables belong to the class, not individual objects.

8. Best Practices for Beginners

- Use meaningful class names.
- Keep fields private and use getters/setters (later lessons).
- Use CamelCase for naming.

9. Hands-On Exercises

1. Create a Book class with title, author, price, and display method.
2. Create a BankAccount class with deposit method.
3. Create a Laptop class and print details of two objects.

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4. Create a Movie class with rating logic.
5. Create a Circle class with area calculation.

10. MCQs for Concept Check

- Q1: Objects store values for fields. (Answer: B)
- Q2: new allocates memory and returns a reference. (Answer: C)
- Q3: One class can create many objects. (Answer: B)
- Q4: Output is 0 because t2.x is default value. (Answer: A)
- Q5: Fields are stored in heap memory. (Answer: B)